

Math 300 Quiz 5 Spring 2024

Key

1. (8 pts) Use mathematical induction to prove that $8^n - 3^n$ is a multiple of 5 for all integers $n \geq 0$

Let $P(n)$ represent the statement $8^n - 3^n = 5x$ for some $x \in \mathbb{Z}$.

Base Case $P(0)$ is true since $8^0 - 3^0 = 1 - 1 = 0 = 5 \cdot 0$.

Induction Step Let $P(k)$ be true for $k \geq 0$ so we have

$$8^k - 3^k = 5x \text{ for some } x \text{ in } \mathbb{Z}. \text{ So } 8^k = 5x + 3^k$$

$$\begin{aligned} \text{This means } 8^{k+1} - 3^{k+1} &= 8 \cdot 8^k - 3 \cdot 3^k = 8(5x + 3^k) - 3 \cdot 3^k \\ &= 40x + 8 \cdot 3^k - 3 \cdot 3^k = 40x + 5 \cdot 3^k = 5(8x + 3^k) = 5y. \end{aligned}$$

Therefore $P(k+1)$ is true whenever $P(k)$ is true

Since $y = 8x + 3^k$ is an integer by closure.

Conclusion Since $P(0)$ is true and $P(k+1)$ is true

whenever $P(k)$ is true and $k \geq 0$ we know by math induction that $P(n)$ is true for all integers $n \geq 0$.

Over $\rightarrow$

2. (4 pts) Give an example of a function from $\{1, 2, 3, 4\}$ to $\{1, 2, 3\}$ that is neither one-to-one nor onto.

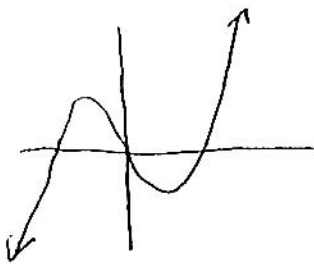
Many examples exist

$$\{(1, 1), (2, 1), (3, 1), (4, 1)\}$$

There must be exactly 4 ordered pairs. Each of 1, 2, 3, and 4 must be the first coordinates exactly once. The second coordinates must be either 1 or 2 or 3 but at least one of these must be left out.

3. (4 pts) Draw a graph or give a rule $[y = f(x)]$ to define a function with domain $\mathbf{R}$ and range $\mathbf{R}$ that is onto but is not one-to-one.

Many examples exist. The graph must fail the horizontal line test but pass the vertical line test.



$$f(x) = x^3 - x \text{ will work}$$

4. (4 pts) State in precise mathematical language what it means for a function f to map set X onto set Y .

For all $y \in Y$, there exist at least one $x \in X$ such that $f(x) = y$

$$\forall y \in Y, \exists x \in X \text{ such that } f(x) = y.$$